Lesson 5: Probability Spaces, Random Variables, and Expectation

AI/ML Mathematics

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1 Motivation and Overview

This lesson develops the core probabilistic tools that underpin every convergence proof in reinforcement learning: conditional probability and Bayes' theorem, expectation defined rigorously via the Lebesgue integral, linearity of expectation, the common discrete distributions (Bernoulli, categorical, geometric), and joint/marginal distributions.

2 Key Definitions (Review from Lesson 4)

We recall the core definitions from Lesson 4 to fix notation. Throughout, $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space (i.e., a measure space with $\mathbb{P}(\Omega) = 1$), and a *random variable* is a measurable function $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with CDF $F_X(x) = \mathbb{P}(X \leq x)$.

The basic properties of probability (complement rule, monotonicity, inclusion-exclusion, union bound, continuity from above/below) and the CDF properties (non-decreasing, correct limits at $\pm\infty$, right-continuity) were proved in Lesson 4. We use them freely below.

3 Conditional Probability and Bayes' Theorem

Definition 3.1 (Conditional Probability)**.** For events $A, B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, the *conditional probability of A given B* is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \tag{1}$$

Proposition 3.2 (Conditional Probability is a Probability Measure)**.** *For fixed B with $\mathbb{P}(B) > 0$, the function $A \mapsto \mathbb{P}(A \mid B)$ is a probability measure on $(\Omega, \mathcal{F})$.*

Proof. We verify the three Kolmogorov axioms.

(K1) $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B) \geq 0$.

(K2) $\mathbb{P}(\Omega \mid B) = \mathbb{P}(\Omega \cap B)/\mathbb{P}(B) = \mathbb{P}(B)/\mathbb{P}(B) = 1$.

(K3) For pairwise disjoint $A_1, A_2, \ldots$:

$$\mathbb{P}\left(\bigcup_n A_n \,\middle|\, B\right) = \frac{\mathbb{P}((\bigcup_n A_n) \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(\bigcup_n (A_n \cap B))}{\mathbb{P}(B)}. \tag{2}$$

The sets $A_n \cap B$ are pairwise disjoint (since the A_n are), so by countable additivity of $\mathbb{P}$:

$$= \frac{\sum_n \mathbb{P}(A_n \cap B)}{\mathbb{P}(B)} = \sum_n \frac{\mathbb{P}(A_n \cap B)}{\mathbb{P}(B)} = \sum_n \mathbb{P}(A_n \mid B). \qquad \square$$

Theorem 3.3 (Chain Rule of Probability)**.** *For events $A_1, \ldots, A_n$ with $\mathbb{P}(A_1 \cap \cdots \cap A_{n-1}) > 0$:*

$$\mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1)\,\mathbb{P}(A_3 \mid A_1 \cap A_2) \cdots \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}). \tag{3}$$

Proof. We proceed by induction on n.

Base case ($n = 2$). By definition of conditional probability, $\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1) \cdot \mathbb{P}(A_2 \mid A_1)$.

Inductive step. Assume the formula holds for $n-1$ events. Let $B = A_1 \cap \cdots \cap A_{n-1}$. By the base case applied to B and A_n:

$$\mathbb{P}(B \cap A_n) = \mathbb{P}(B) \cdot \mathbb{P}(A_n \mid B).$$

By the inductive hypothesis, $\mathbb{P}(B)$ expands as

$$\mathbb{P}(B) = \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1) \cdots \mathbb{P}(A_{n-1} \mid A_1 \cap \cdots \cap A_{n-2}).$$

Combining gives the formula for n events. $\square$

Connection to RL

The chain rule decomposes trajectory probabilities:

$$\mathbb{P}_\pi(S_0, A_0, R_1, S_1, \ldots) = p_0(S_0) \prod_{t=0}^{T-1} \pi(A_t \mid S_t)\, p(S_{t+1}, R_{t+1} \mid S_t, A_t).$$

This factorization is the starting point of importance sampling ratios in off-policy methods (TRPO, PPO).

Law of Total Probability

Theorem 3.4 (Law of Total Probability). *Let $B_1, B_2, \ldots, B_n$ be a partition of Ω (pairwise disjoint, $\bigcup_i B_i = \Omega$) with $\mathbb{P}(B_i) > 0$ for all i. Then for any event $A \in \mathcal{F}$:*

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\, \mathbb{P}(B_i). \tag{4}$$

Proof. Since $\{B_i\}$ partitions Ω:

$$A = A \cap \Omega = A \cap \bigcup_{i=1}^{n} B_i = \bigcup_{i=1}^{n} (A \cap B_i).$$

The sets $A \cap B_1, \ldots, A \cap B_n$ are pairwise disjoint (inherited from the B_i). By finite additivity:

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \cap B_i) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\, \mathbb{P}(B_i),$$

where the last step uses $\mathbb{P}(A \cap B_i) = \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$ from the definition of conditional probability. □

Bayes' Theorem

Theorem 3.5 (Bayes' Theorem). *Let $B_1, \ldots, B_n$ partition Ω with $\mathbb{P}(B_i) > 0$, and let A be an event with $\mathbb{P}(A) > 0$. Then for each j:*

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\, \mathbb{P}(B_j)}{\sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\, \mathbb{P}(B_i)}. \tag{5}$$

Proof. By the definition of conditional probability:

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \cap B_j)}{\mathbb{P}(A)}.$$

The numerator satisfies $\mathbb{P}(A \cap B_j) = \mathbb{P}(A \mid B_j)\mathbb{P}(B_j)$ (definition of conditional probability). The denominator is expanded using the law of total probability (Theorem 3.4): $\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$. Substituting gives the result. □

4 Expectation via the Lebesgue Integral

4.1 Integral of Simple Functions

Definition 4.1 (Integral of a Non-negative Simple Function). Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $\varphi = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}$ be a non-negative simple function in standard representation ($a_i \geq 0$

distinct, $A_i = \varphi^{-1}(\{a_i\})$). The *Lebesgue integral* of φ is

$$\int_\Omega \varphi \, d\mu = \sum_{i=1}^{n} a_i \, \mu(A_i). \tag{6}$$

Proposition 4.2 (Properties of the Simple Integral)**.** *For non-negative simple functions φ, ψ and $c \geq 0$:*

(a) $\int c\varphi \, d\mu = c \int \varphi \, d\mu$.
(b) $\int (\varphi + \psi) \, d\mu = \int \varphi \, d\mu + \int \psi \, d\mu$.
(c) If $\varphi \leq \psi$ pointwise, then $\int \varphi \, d\mu \leq \int \psi \, d\mu$.

Proof. **(a)** If $\varphi = \sum a_i \mathbf{1}_{A_i}$, then $c\varphi = \sum (ca_i)\mathbf{1}_{A_i}$. Hence $\int c\varphi \, d\mu = \sum ca_i \, \mu(A_i) = c \sum a_i \, \mu(A_i) = c \int \varphi \, d\mu$.

(b) Let $\varphi = \sum_{i=1}^{m} a_i \mathbf{1}_{A_i}$ and $\psi = \sum_{j=1}^{n} b_j \mathbf{1}_{B_j}$ in standard representations. The sets $C_{ij} = A_i \cap B_j$ form a partition of Ω (since both $\{A_i\}$ and $\{B_j\}$ are partitions). On C_{ij}, $\varphi + \psi = a_i + b_j$. Therefore:

$$\begin{aligned} \int (\varphi + \psi) \, d\mu &= \sum_{i,j} (a_i + b_j) \, \mu(C_{ij}) \\ &= \sum_i a_i \sum_j \mu(C_{ij}) + \sum_j b_j \sum_i \mu(C_{ij}) \\ &= \sum_i a_i \, \mu(A_i) + \sum_j b_j \, \mu(B_j) = \int \varphi \, d\mu + \int \psi \, d\mu. \end{aligned} \tag{7}$$

Here we used $\sum_j \mu(C_{ij}) = \mu(A_i)$ (since $\bigsqcup_j C_{ij} = A_i$).

(c) $\psi - \varphi \geq 0$, so $\int \psi \, d\mu = \int \varphi \, d\mu + \int (\psi - \varphi) \, d\mu \geq \int \varphi \, d\mu$. □

4.2 Non-negative and General Integrals

Definition 4.3 (Lebesgue Integral – Non-negative Case)**.** For a non-negative measurable $f : (\Omega, \mathcal{F}) \to [0, \infty]$:

$$\int_\Omega f \, d\mu = \sup \left\{ \int_\Omega \varphi \, d\mu : 0 \leq \varphi \leq f, \ \varphi \text{ simple} \right\}. \tag{8}$$

By the simple function approximation theorem (Lesson 3), simple $\varphi_n \uparrow f$ exist, and the monotone convergence theorem gives $\int f \, d\mu = \lim_n \int \varphi_n \, d\mu$.

Definition 4.4 (Lebesgue Integral – General Case)**.** For measurable $f : \Omega \to [-\infty, \infty]$, write $f = f^+ - f^-$ where $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$. Define $\int f \, d\mu = \int f^+ \, d\mu - \int f^- \, d\mu$ whenever at least one side is finite. If both are finite, f is *integrable* ($f \in L^1(\mu)$).

Definition 4.5 (Expectation)**.** Let X be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$. The *expectation* of X is

$$\mathbb{E}[X] = \int_\Omega X \, d\mathbb{P}, \tag{9}$$

provided the integral exists (i.e., $\mathbb{E}[|X|] < \infty$).

Remark 4.6. For a discrete random variable taking values $\{x_k\}_{k \geq 1}$ with $p_k = \mathbb{P}(X = x_k)$, the Lebesgue integral reduces to $\mathbb{E}[X] = \sum_k x_k p_k$ (provided $\sum_k |x_k| p_k < \infty$). This is because $X = \lim_N \sum_{k=1}^{N} x_k \mathbf{1}_{\{X = x_k\}}$ and the integral of each simple approximation is $\sum_{k=1}^{N} x_k p_k$.

5 Linearity of Expectation

Linearity of Expectation

Theorem 5.1 (Linearity of Expectation). *For random variables X and Y with finite expectations, and constants $a, b \in \mathbb{R}$:*

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y]. \tag{10}$$

This holds regardless of whether X and Y are independent.

Proof. We prove linearity in stages, building from simple functions to the general case.

Step 1: Homogeneity, $\mathbb{E}[aX] = a\mathbb{E}[X]$. For $a \geq 0$ and $X \geq 0$: if $\varphi \leq X$ is simple, then $a\varphi \leq aX$ is simple and $\int a\varphi = a \int \varphi$ (Proposition 4.2(a)). Taking the supremum: $\int aX = a \int X$. For general a and X, decompose using positive and negative parts and consider the cases $a \geq 0$ and $a < 0$ separately.

Step 2: Additivity for non-negative simple functions. This is Proposition 4.2(b).

Step 3: Additivity for non-negative measurable functions. Let $X, Y \geq 0$ be measurable. By the simple function approximation theorem, there exist simple $\varphi_n \uparrow X$ and $\psi_n \uparrow Y$. Then $\varphi_n + \psi_n \uparrow X + Y$ and

$$\int (X+Y)\,d\mathbb{P} = \lim_n \int (\varphi_n + \psi_n)\,d\mathbb{P} = \lim_n \left(\int \varphi_n\,d\mathbb{P} + \int \psi_n\,d\mathbb{P} \right) = \int X\,d\mathbb{P} + \int Y\,d\mathbb{P},$$

where the first equality uses the monotone convergence theorem (the integrals of an increasing sequence of non-negative functions converge to the integral of the limit).

Step 4: General case. Write $X = X^+ - X^-$ and $Y = Y^+ - Y^-$. Then $X + Y = (X^+ + Y^+) - (X^- + Y^-)$. By Step 3 applied to the non-negative functions $X^+ + Y^+$ and $X^- + Y^-$: $\int (X^+ + Y^+) = \int X^+ + \int Y^+$ and similarly for the negative parts. It follows that $\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.

Combining: $\mathbb{E}[aX + bY] = \mathbb{E}[aX] + \mathbb{E}[bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$. □

Corollary 5.2 (Linearity for Finite Sums). *For random variables $X_1, \ldots, X_n$ with finite expectations and constants $a_1, \ldots, a_n \in \mathbb{R}$:*

$$\mathbb{E}\left[\sum_{k=1}^{n} a_k X_k\right] = \sum_{k=1}^{n} a_k\,\mathbb{E}[X_k]. \tag{11}$$

Proof. Induction on n. The base case $n = 1$ is homogeneity. For the inductive step, let $Z = \sum_{k=1}^{n-1} a_k X_k$. By Theorem 5.1:

$$\mathbb{E}\left[\sum_{k=1}^{n} a_k X_k\right] = \mathbb{E}[Z + a_n X_n] = \mathbb{E}[Z] + a_n\mathbb{E}[X_n] = \sum_{k=1}^{n-1} a_k\mathbb{E}[X_k] + a_n\mathbb{E}[X_n] = \sum_{k=1}^{n} a_k\mathbb{E}[X_k],$$

where the third equality uses the inductive hypothesis. □

Connection to RL

Linearity is the first step of every Bellman equation derivation. The return $G_t = R_{t+1} + \gamma G_{t+1}$ gives

$$V^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s] = \mathbb{E}_\pi[R_{t+1} \mid S_t = s] + \gamma\,\mathbb{E}_\pi[G_{t+1} \mid S_t = s].$$

The second term is then expanded using the law of total expectation (developed in Lesson 6) to obtain the full Bellman equation.

6 Common Discrete Distributions

Definition 6.1 (Bernoulli Distribution)**.** A random variable X has a *Bernoulli distribution* with parameter $p \in [0,1]$, written $X \sim \text{Bernoulli}(p)$, if $\mathbb{P}(X=1)=p$ and $\mathbb{P}(X=0)=1-p$.

Proposition 6.2 (Expectation of Bernoulli)**.** *If $X \sim \text{Bernoulli}(p)$, then $\mathbb{E}[X]=p$.*

Proof. $\mathbb{E}[X] = 1 \cdot \mathbb{P}(X=1) + 0 \cdot \mathbb{P}(X=0) = p$. □

Definition 6.3 (Categorical Distribution)**.** A random variable X taking values in $\{1,2,\ldots,k\}$ has a *categorical distribution* with parameters $(p_1,\ldots,p_k)$, where $p_i \geq 0$ and $\sum_{i=1}^k p_i = 1$, if $\mathbb{P}(X=i)=p_i$ for each i.

Proposition 6.4 (Expectation of Categorical)**.** *If X has a categorical distribution with parameters $(p_1,\ldots,p_k)$, then $\mathbb{E}[X]=\sum_{i=1}^k i\,p_i$.*

Proof. $\mathbb{E}[X] = \sum_{i=1}^k i \cdot \mathbb{P}(X=i) = \sum_{i=1}^k i\,p_i$. □

Connection to RL

The categorical distribution models action selection under a stochastic policy: $A_t \sim \pi(\cdot \mid s)$ is categorical with $\mathbb{P}(A_t = a) = \pi(a \mid s)$. The policy gradient theorem computes $\nabla_\theta \mathbb{E}[R]$ by differentiating through these categorical probabilities.

Definition 6.5 (Geometric Distribution)**.** $X \sim \text{Geometric}(p)$ for $p \in (0,1]$ if $\mathbb{P}(X=k) = (1-p)^{k-1}p$ for $k=1,2,3,\ldots$.

Proposition 6.6 (Expectation of Geometric)**.** *If $X \sim \text{Geometric}(p)$, then $\mathbb{E}[X]=1/p$.*

Proof. We compute:

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} k\,(1-p)^{k-1}p = p\sum_{k=1}^{\infty} k\,(1-p)^{k-1}. \tag{12}$$

Let $q=1-p$. We evaluate $\sum_{k=1}^{\infty} kq^{k-1}$ by differentiating the geometric series:

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q} \quad \Longrightarrow \quad \frac{d}{dq}\sum_{k=0}^{\infty} q^k = \sum_{k=1}^{\infty} kq^{k-1} = \frac{1}{(1-q)^2}.$$

The differentiation is justified because the power series converges uniformly on $[0,q_0]$ for any $q_0<1$, and $q=1-p\in[0,1)$. Substituting $1-q=p$:

$$\mathbb{E}[X] = p \cdot \frac{1}{p^2} = \frac{1}{p}.$$ □

Remark 6.7. The geometric distribution models the number of trials until the first success. In RL, it arises in episode length analysis: if each step has termination probability p, the episode length is $\text{Geometric}(p)$ with expected length $1/p$.

7 Joint and Marginal Distributions

Definition 7.1 (Joint Distribution)**.** For random variables X and Y on $(\Omega,\mathcal{F},\mathbb{P})$, the *joint distribution* is the probability measure $\mu_{X,Y}$ on $(\mathbb{R}^2,\mathcal{B}(\mathbb{R}^2))$ defined by

$$\mu_{X,Y}(B) = \mathbb{P}\big((X,Y)\in B\big) \quad \text{for all } B \in \mathcal{B}(\mathbb{R}^2). \tag{13}$$

In the discrete case, the *joint PMF* is $p_{X,Y}(x,y) = \mathbb{P}(X=x, Y=y)$.

Definition 7.2 (Marginal Distribution)**.** The *marginal PMF* of X is obtained by summing the joint PMF over all values of Y:

$$p_X(x) = \sum_y p_{X,Y}(x, y). \tag{14}$$

Proposition 7.3 (Marginalization via Total Probability)**.** *Let X and Y be discrete random variables. Then for every x in the range of X:*

$$p_X(x) = \sum_y p_{X,Y}(x, y). \tag{15}$$

Proof. Let Y take values in $\{y_1, y_2, \ldots\}$. The events $B_j = \{Y = y_j\}$ are pairwise disjoint with $\bigcup_j B_j = \Omega$. For $A = \{X = x\}$: $A = \bigsqcup_j (A \cap B_j)$, so by countable additivity $p_X(x) = \sum_j \mathbb{P}(X = x, Y = y_j) = \sum_j p_{X,Y}(x, y_j)$. □

8 Basic Properties of Expectation

Proposition 8.1 (Monotonicity of Expectation)**.** *If $X \leq Y$ almost surely and both expectations exist, then $\mathbb{E}[X] \leq \mathbb{E}[Y]$.*

Proof. $Y - X \geq 0$ a.s., so $\mathbb{E}[Y - X] \geq 0$ (the integral of a non-negative function is non-negative). By linearity: $\mathbb{E}[Y] - \mathbb{E}[X] = \mathbb{E}[Y - X] \geq 0$. □

Proposition 8.2 (Triangle Inequality for Expectation)**.** *If $\mathbb{E}[|X|] < \infty$, then $|\mathbb{E}[X]| \leq \mathbb{E}[|X|]$.*

Proof. Since $-|X| \leq X \leq |X|$, monotonicity gives $-\mathbb{E}[|X|] \leq \mathbb{E}[X] \leq \mathbb{E}[|X|]$, which is $|\mathbb{E}[X]| \leq \mathbb{E}[|X|]$. □

Proposition 8.3 (Expectation of an Indicator)**.** *For any event $A \in \mathcal{F}$, $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$.*

Proof. $\mathbf{1}_A$ is a simple function with standard representation $1 \cdot \mathbf{1}_A + 0 \cdot \mathbf{1}_{A^c}$. By Definition 4.1: $\mathbb{E}[\mathbf{1}_A] = 1 \cdot \mathbb{P}(A) + 0 \cdot \mathbb{P}(A^c) = \mathbb{P}(A)$. □

9 Exercises

Exercise 9.1. Extend the law of total probability (Theorem 3.4) to countable partitions: if $B_1, B_2, \ldots$ is a countable partition of Ω with $\mathbb{P}(B_i) > 0$, show that $\mathbb{P}(A) = \sum_{i=1}^{\infty} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$.

Exercise 9.2. In a three-state MDP, the agent starts in state s_1, s_2, or s_3 with prior probabilities 0.5, 0.3, 0.2. The probability of receiving a positive reward ($R > 0$) from each state is 0.9, 0.4, 0.1 respectively. Given that a positive reward was observed, compute the posterior probability of each starting state using Bayes' theorem.

Exercise 9.3. Let $G_t = \sum_{k=0}^{T-t-1} \gamma^k R_{t+k+1}$ be the discounted return in a finite-horizon MDP with horizon T and discount factor $\gamma \in (0, 1)$. Using linearity of expectation, show that $\mathbb{E}[G_t] = \sum_{k=0}^{T-t-1} \gamma^k \mathbb{E}[R_{t+k+1}]$.

Exercise 9.4. Prove the *memoryless property* of the geometric distribution: if $X \sim \text{Geometric}(p)$, then for all $m, n \geq 1$:

$$\mathbb{P}(X > m + n \mid X > m) = \mathbb{P}(X > n).$$

Hint: Compute $\mathbb{P}(X > k) = (1 - p)^k$ and use the definition of conditional probability.

Exercise 9.5. Let X be a non-negative integer-valued random variable with CDF F_X. Show that $\mathbb{E}[X] = \sum_{k=0}^{\infty} \mathbb{P}(X > k) = \sum_{k=0}^{\infty}(1 - F_X(k))$. *Hint:* Write $X = \sum_{k=0}^{\infty} \mathbf{1}_{\{X>k\}}$ and apply linearity.

Exercise 9.6. For $A \in \mathcal{F}$, compute $\mathbb{E}[\mathbf{1}_A^2]$ and use it to find $\mathbb{E}[\mathbf{1}_A^2] - (\mathbb{E}[\mathbf{1}_A])^2$. The quantity you obtain is the variance of a Bernoulli random variable (to be formally defined in Lesson 7).

Exercise 9.7. Let (X, Y) have joint PMF $p_{X,Y}(0,0) = 1/4$, $p_{X,Y}(0,1) = 1/4$, $p_{X,Y}(1,0) = 1/4$, $p_{X,Y}(1,1) = 1/4$. Compute the marginal PMFs p_X and p_Y, verify $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, and compute $\mathbb{E}[X + 2Y]$ using linearity.

10 Summary

Result	**Statement**	**Role**
Conditional probability	$\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$	Defines Bayesian updates
Chain rule	$\mathbb{P}(\bigcap A_i) = \prod \mathbb{P}(A_i \mid \bigcap_{j<i} A_j)$	Trajectory probabilities
Total probability	$\mathbb{P}(A) = \sum \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$	Marginalizes over partitions
Bayes' theorem	$\mathbb{P}(B_j \mid A) \propto \mathbb{P}(A \mid B_j)\mathbb{P}(B_j)$	Posterior updates
Expectation	$\mathbb{E}[X] = \int X \, d\mathbb{P}$ via Lebesgue integral	Value functions
Linearity	$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$	Bellman equation derivations
Bernoulli	$\mathbb{E}[X] = p$	Binary outcomes
Categorical	$\mathbb{E}[X] = \sum i \, p_i$	Policy action selection
Geometric	$\mathbb{E}[X] = 1/p$	Episode length
Marginalization	$p_X(x) = \sum_y p_{X,Y}(x, y)$	Summing over nuisance variables

Next lesson: Conditional Expectation and the Tower Property – the rigorous definition of $\mathbb{E}[Y \mid X]$ as a random variable, its key properties (linearity, pull-out, tower), and applications to the Bellman equation.

References